

Quarto Grid Challenge

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1 Introduction

Long bones develop through the process of Endochondral Ossification (EO) (Kronenberg 2003; Aghajanian and Mohan 2018; Nguyen et al. 2017). This process starts with the condensation of mesenchymal cells, continuing with intricate processes of differentiation, proliferation, hypertrophy, apoptosis and matrix modification (Kronenberg 2003; Aghajanian and Mohan 2018; Nguyen et al. 2017). During this transformation, hyaline cartilage is replaced by bone, developing primary (POC) and secondary (SOC) ossification centers; the former allows longitudinal development, while the latter promotes epiphysis formation. This progression is modulated by multiple factors, including cellular, biochemical, and mechanical effects (Carter and Wong 1988; Sundaramurthy and Mao 2006). Disturbances on this growth mechanism can cause different pathologies in children and teenagers; therefore, understanding this process is crucial to explain, prevent and treat these disorders (Comellas and Shefelbine 2022; Ogden, Hensinger, and McCollough 2000). For this reason, several mathematical and computational models have been proposed to simulate EO and SOC emergence. Early works suggested that cyclical mechanical stimuli could influence ossification, allowing the prediction on location of SOC (Carter and Wong 1988).

Additionally, other factors have been considered, e.g., bone remodelling (Wong and Carter 1990; Stevens, Beaupré, and Carter 1999), biochemical feedback loops and cellular mechanisms (D. A. Garzón-Alvarado, García-Aznar, and Doblaré 2009; Diego Alexander Garzón-Alvarado, Garca-Aznar, and Doblaré 2009; Stoop, Yokoyama, and Adachi 2025), or a combination of these elements

(Peinado Cortés, Vanegas Acosta, and Garzón Alvarado 2011; Vaca-González et al. 2018; Carrera-Pinzón et al. 2020; Sadeghian, Shapiro, and Shefelbine 2021). Given that multiple SOC can appear on a single epiphysis (Ogden, Hensinger, and McCollough 2000), studies based on reaction-diffusion modeling have been conducted to analyze the influence of epiphyseal size on the number of SOC (Diego Alexander Garzón-Alvarado 2011). However, previous models have mostly focused on the emergence of a single SOC, overlooking multiple SOC presence on the epiphysis, as observed in the humerus, tibia, femur, and other bones. Therefore, this study builds on previous work that predicted SOC locations based primarily on geometric and mechanical factors (Bustamante-Porras et al. 2025) by incorporating biochemical influences. It aims to explore how geometry, mechanics, and biochemical conditions interact to govern SOC formation, including double-center ossification cases.

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